Open problems:
Classical uncertainty principle

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

No item below blocks the verification of a main claim of the brief.

From the brief

1. **The quantum case.** The conclusion says that "in general, the uncertainty principle is a consequence of the entropic lower bound", since pure quantum states have zero entropy. The classical bound $\hbar/e \approx 0.37\,\hbar$ is weaker than Heisenberg's $\hbar/2$, and no entropic derivation of the quantum bound is given. Whether, and in what form, the quantum principle follows from the zero entropy of pure states is open. (Brief, Sec. 3.)

2. **The equality case of eq. (5).** The brief says when equality holds in eq. (3) (see the log, "Problems found", first item) but not in eq. (5). Equality in Gibbs' inequality would require $\rho = |\{f,g\}|\,(G \circ (f,g))$ almost everywhere, with G a product of Gaussians, and then also that (f,g) is onto up to a null set. The exact conditions are not stated. (Brief, eq. (5).)

From the formalization

1. **Several degrees of freedom.** ZENT and CUP are formalized for one DOF, on a density on $\mathbb{R}^2$. That the marginal of a density on $(\mathbb{R}^2)^n$ is again an ensemble with finite entropy and finite second moments is not formalized. (Log, "Differences", second item.)

2. **Entropy in new variables.** The brief's identity $S[\rho_{fg}] = S[\rho_{qp}] + E[\ln|\{f,g\}|]$ for the density ρ_{fg} of (f,g) is not formalized. Eq. (5) is proved without forming ρ_{fg}. (Log, "Differences", fourth item.)

3. **The Lagrange computation.** The variational computation of the proof of Theorem 2 is not formalized; Gibbs' inequality replaces it. (Log, "Differences", first item.)